# MILITARY SPECIFICATION
# CARTRIDGES, 14.5MM, ARTILLERY TRAINING, M181A1, M182A1 AND M183A1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers cartridges, 14.5MM, Artillery Training, M181A1 with 3 second delay fuze, M182A1 with 6 second delay fuze and M183A1 with point detonating fuze for use in Trainer, Field Artilley, 14.5MM, M31.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIlitary
- MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
- MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS

Armament Command
- C11740405 - Classification of Cartridge Case Defects
- D11740357 - Cartridge, 14.5MM, Artillery Training, M181A1
- D11740379 - Cartridge, 14.5MM, Artillery Training, M182A1
- D11740384 - Cartridge, 14.5MM, Artillery Training, M183A1
- F11738800 - Packing and Marking, Cartridges, 14.5MM, Artillery Training; Cartons; Box, Ammunition, M2A1 Box, Wirebound



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2.1 (Cont'd)

PUBLICATIONS

AMCR 715-505, Vol. 8 - Ammunition Ballistics Acceptance Test Methods, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridges shall comply with the requirements specified on Drawings D11740357, D11740379 and D11740384, as applicable, referenced specifications and the following:

3.2 Projectile extraction.-The force required to extract the projectile from the cartridge case shall be not less than 3 kilograms (kgs).

3.3 Velocity.-The average velocity of the sample cartridges conditioned at 21.1 degrees ± 1.1 degrees centigrade (C) shall be 100 meters per second (m/sec) plus or minus 3 meters per second at 4 meters (m) from the muzzle. The extreme variation of the velocities shall not exceed 15 meters per second

3.4 Accuracy.-The sum of the extreme spreads (horizontal plus vertical) of the pattern on a target located at 50 meters from the muzzle shall be not greater than 50 centimeters (cm).

3.5 Projectile function.-The M181A1 projectile, when the cartridge is fired at ambient temperature, shall have a delay time of 3 seconds ± 0.45 seconds, and the M182A1 projectile shall have a delay time of 6 seconds ± 0.9 seconds. The M183A1 point detonating projectile shall detonate upon impact with dry earth at a distance of 400 meters. The delay time to smoke emission of the M181A1 and M182A1 projectiles, and the distance at which the M183A1 projectile detonates shall be determined by measurement from the muzzle of weapon. The projectile shall produce a visible puff of smoke in daylight hours when viewed from the gun position.

3.6 Function and casualty.-The cartridge shall function without casualty.

3.7 Projectile sensitivity.-The M183A1 projectile shall not detonate when the cartridge is dropped onto a steel plate from a height of 5 meters.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 130 cartridges. The sample cartridges shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample cartridges shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.Failure of the sample cartridges to comply with requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.



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4.3.1.2 Lot identification.-Each lot of ammunition shall be identified by type, caliber and model, as well as lot number, in accordance with MIL-STD-1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 1.50 percent and the AQL for the minor class shall be 2.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects.-The classification of defects shall be as specified in Table I.

TABLE I

| No. | Defect and Method of Inspection | Critical | Major | Minor |
|---|---|---|---|---|
| | Visual | | | |
| | Cartridge | | | |
| 1 | Discolored, dirty, oily, smeared | | | X |
| 2 | Corroded; or stained, if etched | | X | |
| 3 | Improper crimp (case-projectile) | | X | |
| | Case | | | |
| 4 | Split case | | X | |
| 5 | Perforated case | X | | |
| 6 | Draw scratch | | | X |
| 7 | Bulge | | | X |
| 8 | Illegible or missing head stamp | | | X |
| 9 | Defective mouth | | | X |
| 10 | Defective head | | | X |
| 11 | Marking incorrect, illegible or missing | | | X |
| | Primer | | | |
| 12 | Loose primer | | X | |
| 13 | Inverted primer | | X | |
| 14 | No primer | | X | |
| 15 | Cocked primer | | X | |
| 16 | Waterproofing missing (Joint primer-pocket) | | | X |
| 17 | Nicked or dented primer | | | X |
| 18 | Defective crimp | | | X |
| | Projectile | | | |
| 19 | Split | | | X |
| 20 | Absence of lubricant on periphery | | | X |
| 21 | Loose projectile | | X | |



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TABLE I (Cont'd)

| No. | Defect and Method of Inspection | Critical | Major | Minor |
|---|---|---|---|---|
| | Visual | | | |
| | Gaging | | | |
| 22 | Overall length | | X | |
| 23 | Cartridge profile failure (requiring more than 9.00 kg dead weight to insert cartridge in profile and alignment gage) | | X | |
| 24 | Rim diameter | | X | |
| 25 | Rim thickness | | X | |
| 26 | Depth of primer | | X | |

4.3.3 Tests.-The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

| Test | Number of Cartridges | Requirement Paragraph |
|---|---|---|
| Projectile Extraction 1/ | 10 | 3.2 |
| Velocity 2/ | 32 | 3.3 |
| Accuracy 3/ | - | 3.4 |
| Projectile Function 4/ | - | 3.5 |
| Function and Casualty 5/ | 50 | 3.6 |
| Projectile Sensitivity 6/ | 25 | 3.7 |

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.



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3/ Failure of any target to comply with the applicable requirements shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of any target in the second sample to comply with the applicable requirements shall be cause for rejection of the lot. This test shall be conducted simultaneously with the velocity test.

4/ Failure of seven or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than two but less than seven cartridges fail in the first test, a sacond sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second samples, seven or more cartridges fail to comply with the applicable requirement. This test shall be conducted simultaneously with the function and casualty test.

5/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the accepted number for the cumulative sample in Table III. If the number of defects found in the first tests exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantity specified under function and casualty test shall be fired. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

6/ Failure of one or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. No second sample permitted.

4.3.3.2 Firing defects.-Firing defects and acceptance numbers shall be specified in Table III.

TABLE III

| No. | Defects | Acceptance Numbers (1st Sample) | Acceptance Numbers Cumulative (1st & 2nd Sample) |
|---|---|---|---|
| 1 | Misfire | 1 | 2 |
| 2 | Projectile remaining in bore 1/ | 0 | - |
| 3 | Primer leak | | |
| | a. Perforation in firing pin indent in primer cup | 1 | 2 |



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Table III (Cont'd)

| No. | Defects | Acceptance Numbers (1st Sample) | Acceptance Numbers Cumulative (1st & 2nd Sample) |
|---|---|---|---|
| | b. Escape of gas through primer cup other than 3a | 1 | 2 |
| | c. Escape of gas around primer cup more than 50% of periphery | 4 | 6 |
| | d. Blown primer | 0 | 1 |
| | e. Primer falls out of pocket on retraction of bolt | 0 | 1 |
| | f. Primer remains in pocket but is physically loose | 2 | 3 |
| 4 | Case casualties | | |
| | a. Longitudinal split 2/ | | |
| | (1) Mouth (I) | 2 | 3 |
| | (2) Body (J) | 0 | 2 |
| | (3) Body (K) | 0 | 2 |
| | (4) Body to rim (L) | 0 | 1 |
| | (5) Body through rim (M) | 0 | 1 |
| | b. Circumferential Rupture 2/ | | |
| | (1) Partial body (J) | 2 | 3 |
| | (2) Partial body (K) | 0 | 1 |
| | (3) Partial rim (L) | 0 | 1 |
| | (4) Complete | 0 | 1 |
| 5 | Failure to Extract | 1 | 2 |

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parenthesis, see Drawing C11740405.

4.3.4 Packaging, packing and marking inspection.-During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to determine that they conform to the applicable cartridge drawing. All non-conforming cartridges shall be classed as a major defect and shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to Drawing F11738800.

4.3.5 Inspection equipment.-Except as otherwise provided by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Projectile extraction.-The cartridges shall be tested in an approved projectile extractor machine. The rate of travel of the test head shall be not less than 1.27 millimeters (mm) nor more than 2.54mm per second.



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4.4.2 Velocity.-This test shall be conducted simultaneously with the accuracy test. The lumiline screens for measuring projectile velocity shall be positioned as follows: Distance from gun muzzle to first screen, two meters. Distance between first and second screens, four meters. The barrels used for velocity measurement are to be gaged and the chamber dimension verified prior to placing them in service. The barrel, if measurements are satisfactory, shall be identified as such. All ammunition shall be conditioned at 21.1 degrees ± 1.1 degrees C for at least two hours prior to firing. A constant temperature box shall be placed near the test area. The firing cadence shall be two minutes ± 10 seconds per cartridge. This shall include a dwell time of the cartridge in the warm chamber not to exceed 30 seconds. In the event an abnormal individual result is obtained which can definitely be traced to faulty test technique, the velocity value obtained shall be eliminated from the official report and a replacement cartridge fired. If an abnormal individual result is obtained which cannot definitely be attributed to test technique, the results obtained shall be entered on the official report. If the velocity is not obtained on an individual test round, no official entry shall be recorded, and a replacement cartridge shall be fired.

4.4.3 Accuracy.-This test shall be conducted in accordance with the applicable methods of AMCR 715-505, Volume 8, except the target size shall be approximately 90 cm by 90 cm at a distance of 50 m. Accuracy tests shall consist of firing 16 cartridges at each target.

4.4.4 Projectile function.-Self detonation and smoke cloud emission shall be observed from the gun position and shall be visible in daylight hours. A record shall be made of all M181A1 and M182A1 projectiles failing their three and six second smoke emission delay characteristics respectively, and the M183A1 projectiles failing to detonate and emit smoke upon impact. An informational record shall be made of any abnormal behavior such as erratic flight or muzzle flash of greater intensity than that produced by other projectiles in the test sample.

4.4.5 Function and casualty.-This test shall be conducted simultaneously with the projectile function test. Firing defects shall be recorded in accordance with the terminology listed in Table III.

4.4.6 Projectile sensitivity.-Each M183A1 test cartridge shall be dropped point down, through a nominal one inch diameter tube, positioned perpendicular over a flat rigid steel plate. The cartridge when released through the tube shall fall at normal gravity acceleration.

4.4.7 Defect penalty.-In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.



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5. PREPARATION FOR DELIVERY

5.1 Packing, Level A (Worldwide Shipment).-The cartridges shall be packed in accordance with Drawing F11738800.

5.2 Marking and labeling.-Packing boxes shall be marked and labeled in accordance with the drawing cited in 5.1.

6. NOTES

6.1 Ordering data.-Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Applicable cartridge drawing number (See 3.1).

6.1.3 Type and level of packing.

6.1.4 Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.5 Provisions for the submission of acceptance of inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.6 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Safety precautions.-Fumes produced by functioning of the projectiles are considered toxic and inhalation of such fumes should be avoided. Adequate exhausts should be provided to remove fumes from test areas.

6.3 Hazard notice.-The cartridges described herein and certain of their components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosions, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

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